Princess Sumaya University for Technology
جامعة الأميرة سميّة للتكنولوجيا

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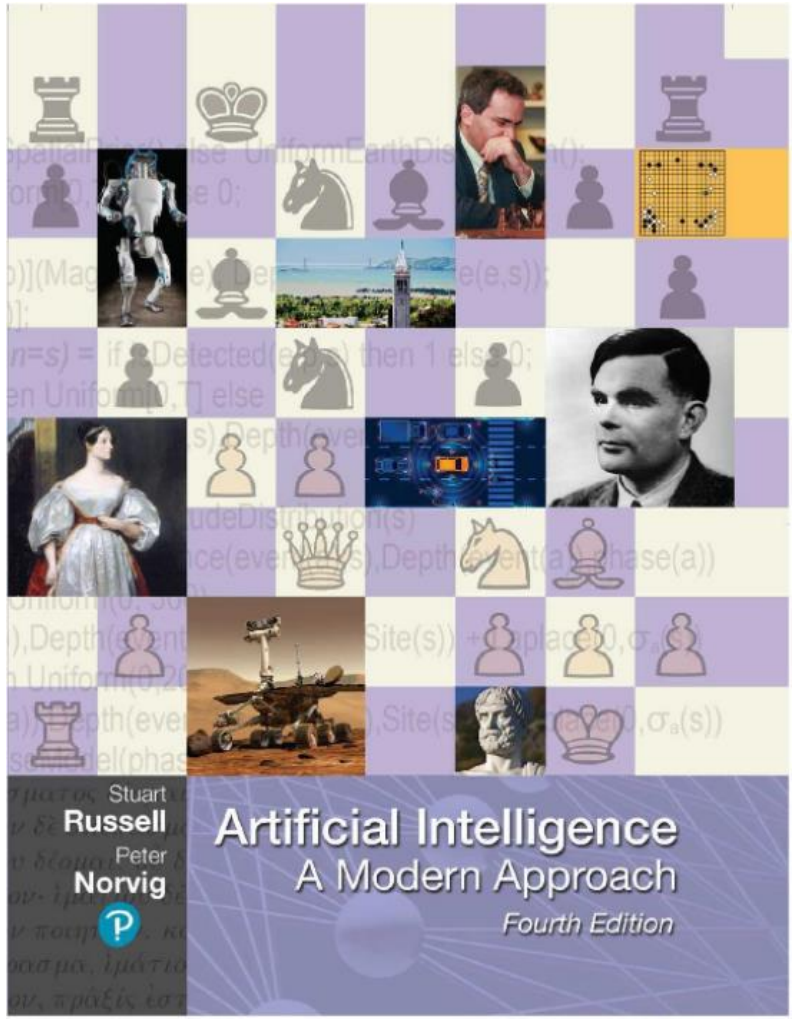
Chapter 1

Introduction to Artificial Intelligence

References



Artificial Intelligence: A Modern Approach, Stuart Russell and Peter Norvig, 4th Edition, 2016, ISBN: 978-0134610993

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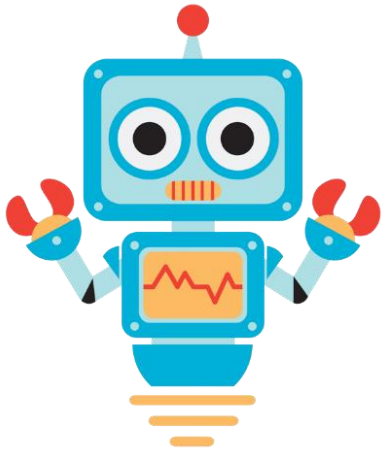
What is Artificial Intelligence?

Artificial → Not Natural



Intelligence → Understand, think, & learn

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What is Artificial Intelligence?

- Intelligence → "the capacity to learn and solve problems" (Websters dictionary)
  - the ability to solve novel problems
  - the ability to act rationally
  - the ability to act like humans

- Artificial Intelligence → build and understand intelligent entities or agents

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What is Artificial Intelligence?

- "The branch of computer science that is concerned with the automation of intelligent behavior." (Luger and Stublefield, 1993)
- AI leverages computers and machines to mimic the problem-solving and decision-making capabilities of the human mind.

Dimensions of Definition		In terms of human performance	In terms of rationality
	Thought Processes and Reasoning	Systems that think like humans	Systems that think rationally
	Behavior	Systems that act like humans	Systems that act rationally

Measure of Success

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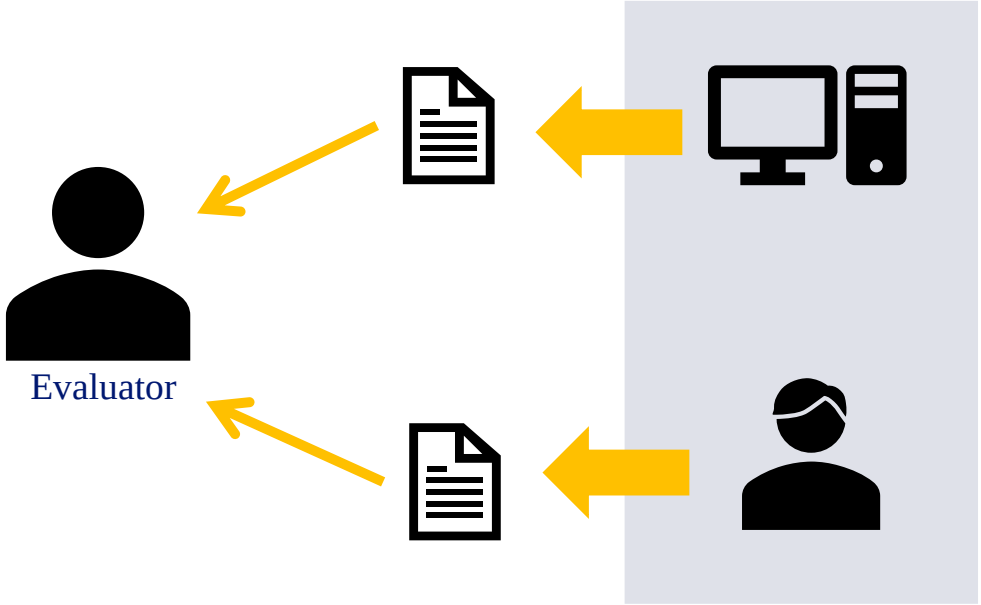
Acting humanly: The Turing Test approach

AI is

"The art of creating machines that perform functions that require intelligence when performed by people" (Kurzweil, 1990)

"The study of how to make computers do things at which, at the moment, people are better." (Rich and Knight, 1991)

- Turing Test → provide a satisfactory operational definition of intelligence
  - A computer passes the test if a human interrogator, after posing some written questions, cannot tell whether the written responses come from a person or from a computer.



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Acting humanly: The Turing Test approach

- The computer would need to possess the following capabilities to pass the Turing Test :
  - Natural language processing → enables it to communicate successfully
  - Knowledge representation → stores what it knows or hears
  - Automated reasoning → use the stored information to answer questions and to draw new conclusions
  - Machine learning → adapt to new circumstances and detects and extrapolate patterns

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Acting humanly: The Turing Test approach

- Total Turing Test → includes a video signal so that the interrogator can test the subject's perceptual abilities, as well as the opportunity for the interrogator to pass physical objects "through the hatch."
  - the interrogation should involve visual acuity and physical interaction

- To pass the total Turing Test, the computer will need:
  - Computer vision → perceive objects
  - Robotics → manipulate objects and move about

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Thinking humanly: Cognitive modeling approach

AI is

"The exciting new effort to make computers think . . . machines with minds, in the full and literal sense." (Haugeland, 1985)

"[The automation of] activities that we associate with human thinking, activities such as decision-making, problem solving, learning . . ." (Bellman, 1978)

We Must Have Some Way Of Determining How Humans Think

- Cognitive Science → Interdisciplinary field (AI, psychology, linguistics, philosophy, anthropology) that tries to construct computational theories of human cognition.
  - Try to get "inside" our minds

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Thinking rationally: Laws of thought approach

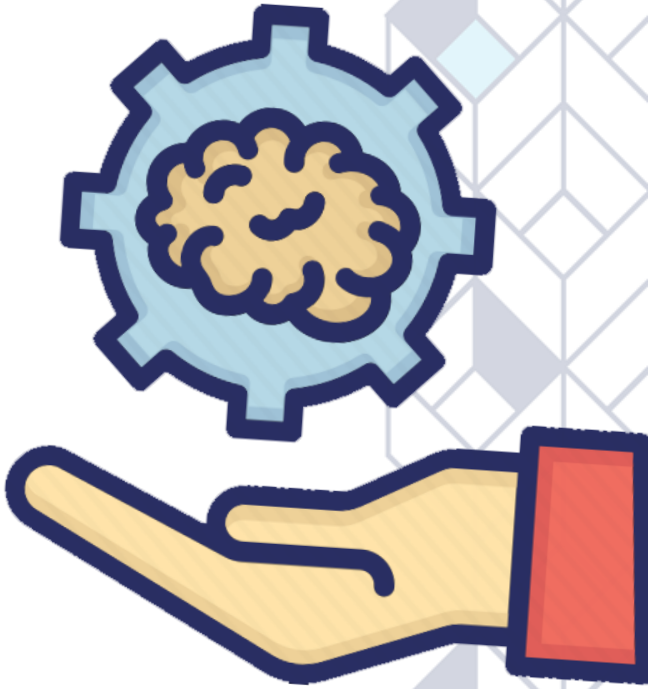
AI is

"The study of mental faculties through the use of computational models." (Charniak and McDermott, 1985)

"The study of the computations that make it possible to perceive, reason, and act." (Winston, 1992)

A system is rational if it does the "right thing," given what it knows

- Represent facts about the world via logic
- Logicist Program → programs could solve any solvable problem described in logical notation.
  - if no solution exists, the program might loop forever



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Acting rationally: Rational agent approach

AI is

"Computational Intelligence is the study of the design of intelligent agents." (Poole et al., 1998)

"AI . . . is concerned with intelligent behavior in artifacts." (Nilsson, 1998)

- An agent → an entity that perceives its environment and is able to execute actions to change it, and create and pursue goals.
- A rational agent → an agent that acts to achieve the best outcome or the best-expected outcome (if there is uncertainty).
- All the skills needed for the Turing Test also allow an agent to act rationally.

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Foundations of Artificial Intelligence

Many older disciplines contribute to a foundation for artificial intelligence

Philosophy	Logic, methods of reasoning, mind as physical system, foundations of learning, language, rationality
Mathematics	Formal representation and proof, algorithms, computation, (un)decidability, (in)tractability
Probability/Statistics	Modeling uncertainty, learning from data
Economics	Utility, decision theory, rational economic agents
Neuroscience	Neurons as information processing units
Psychology/Cognitive Science	How do people behave, perceive, process cognitive information, represent knowledge
Computer engineering	Building fast computers
Control theory	Design systems that maximize an objective function over time
Linguistics	Knowledge representation, grammars

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History of Artificial Intelligence

- 1940-1950: Early beginnings
  - 1943: McCulloch & Pitts: Boolean circuit model of brain
  - 1950: Turing's “Computing Machinery and Intelligence”
- 1950-1970
  - 1950s: Early AI programs, including Samuel's checkers program, Newell & Simon's Logic Theorist, Gelernter's Geometry Engine
  - 1956: Dartmouth meeting: “Artificial Intelligence” adopted
  - 1965: Robinson's complete algorithm for logical reasoning
- 1970-1990: Knowledge-based approaches
  - 1969—79: Early development of knowledge-based systems
  - 1980—88: Expert systems industry booms
  - 1988—93: Expert systems industry busts: “AI Winter”

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History of Artificial Intelligence

- 1990—: Statistical approaches
  - Resurgence of probability, focus on uncertainty
  - General increase in technical depth
  - Agents and learning systems… “AI Spring”?
- 2000—: Where are we now?

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History of Artificial Intelligence

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What can Artificial Intelligence do today?

- Robotic vehicles
- Speech recognition
- Autonomous planning and scheduling
- Game playing
- Spam fighting
- Logistics planning
- Robotics
- Machine Translation

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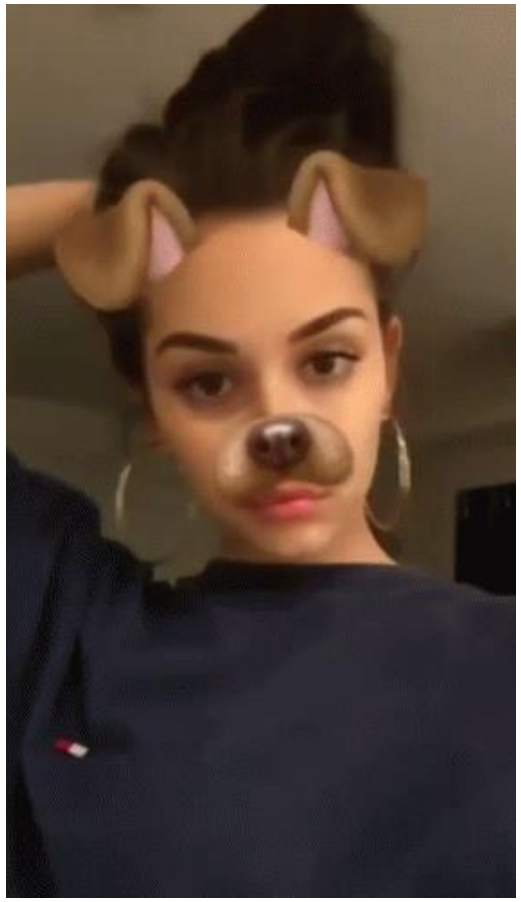
Artificial Intelligence in Your Everyday Life

- Filters



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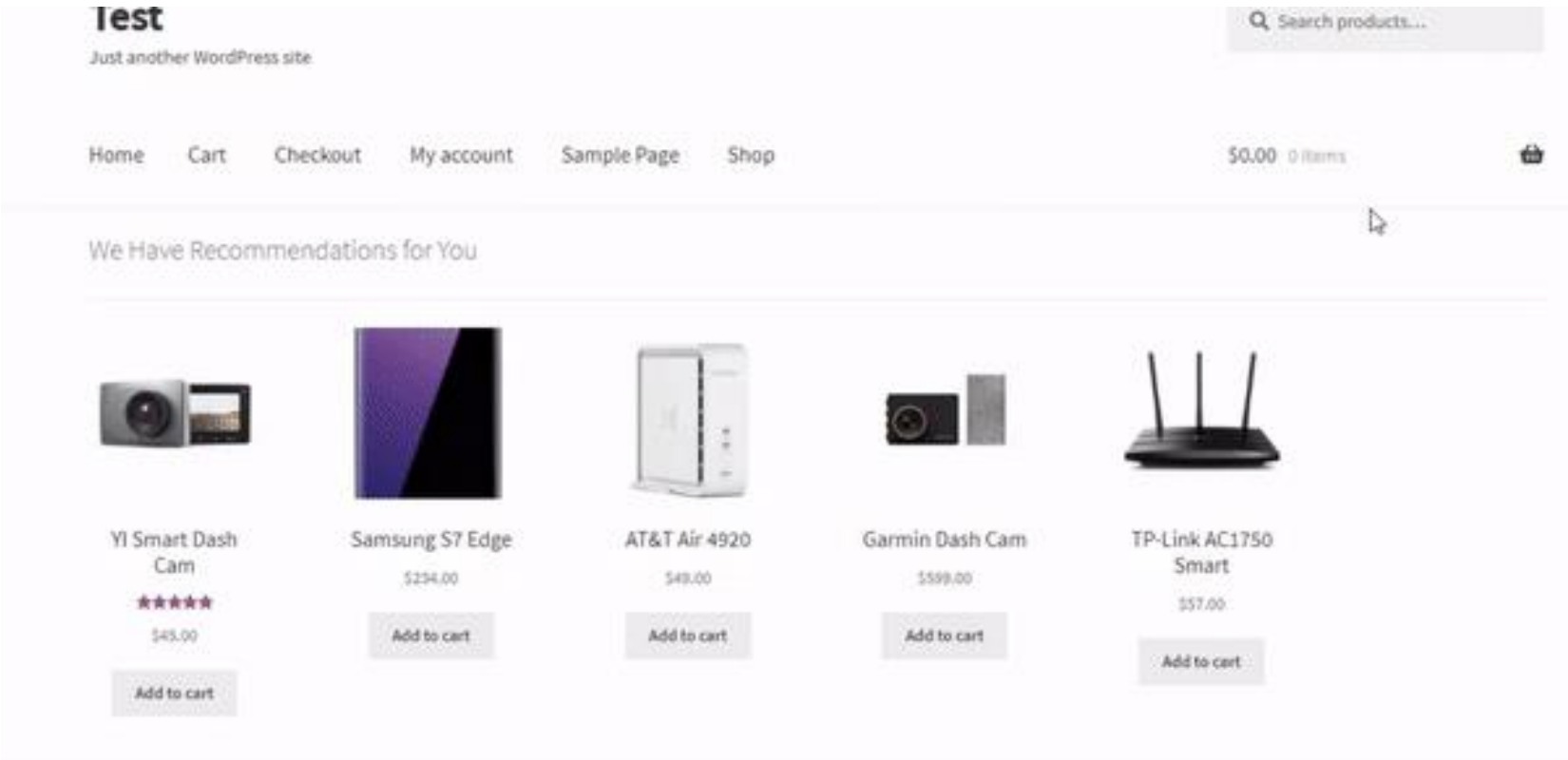
Artificial Intelligence in Your Everyday Life

- Recommendations



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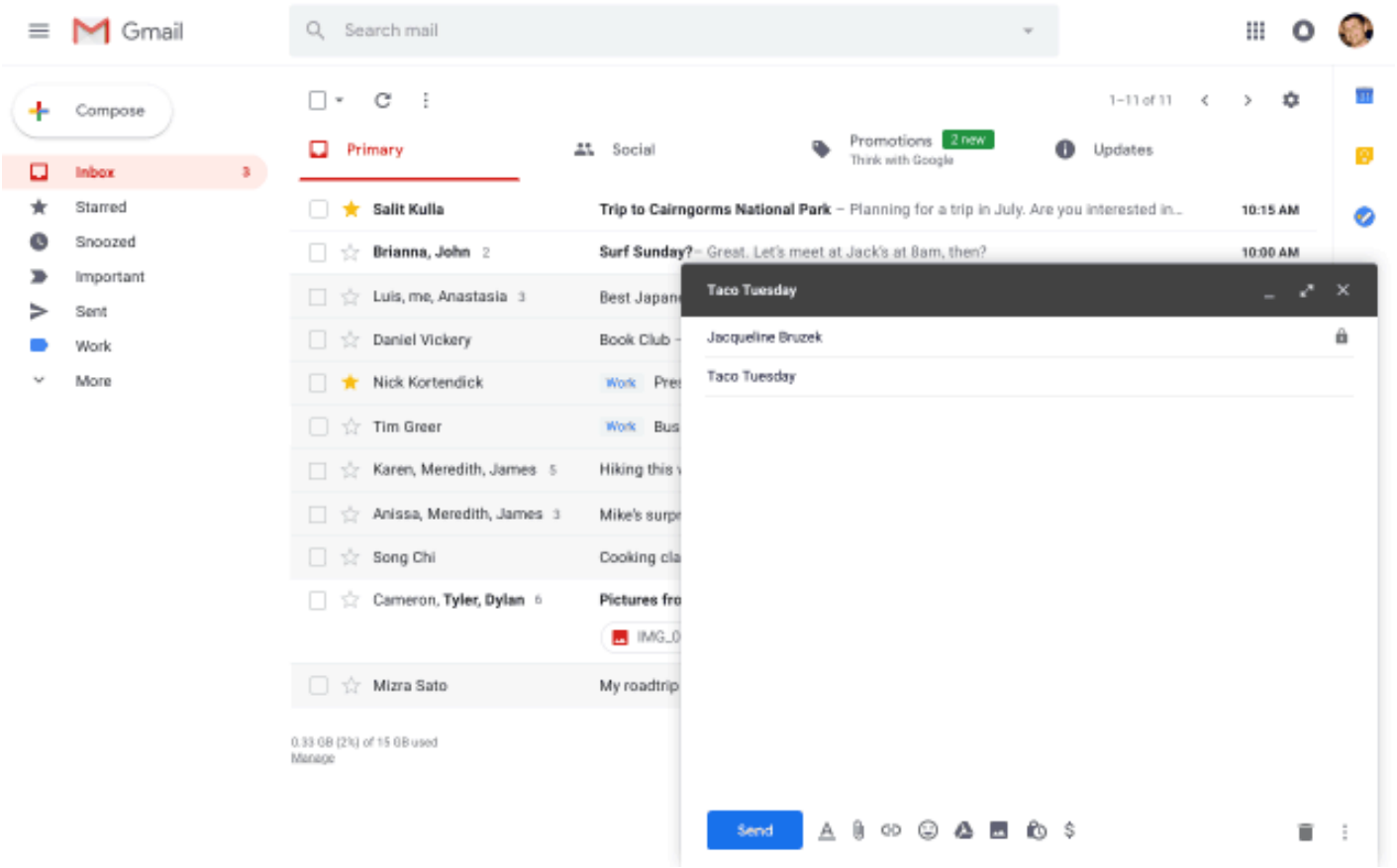
Artificial Intelligence in Your Everyday Life

- Auto-Complete Sentences



Artificial Intelligence in Your Everyday Life

- Virtual Assistant

Google Assistant

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Artificial Intelligence in Your Everyday Life

- Auto-Parking

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Any Question

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